MARWAN ALTAEZI

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Professional Summary:

Chemical Engineering student with industrial training experience at Saudi Aramco and a powerful foundation in process engineering, reactor design, materials engineering, and technical problem-solving. Experienced in chemical reaction engineering, Python-based engineering applications, and industrial operations.

Education:

Bachelor's Degree in Chemical Engineering, King Fahd University of Petroleum & Minerals Graduated: 2026

Concentration: Non-Metallic Materials (Polymers). Academic focus includes polymer chemistry, polymer characterization, process design, fluid mechanics, and engineering problem-solving for industrial and materials-based applications. The program has strengthened my foundation in chemical engineering principles and their practical use in industrial systems.

Experiences:

Material Consulting Service Department Intern, Saudi Aramco Jun 2025 - October 2025

- Completed an internship at Saudi Aramco, gaining practical exposure to industrial operations, professional work environments, and engineering practices.
- Built a stronger understanding of materials-related engineering workflows, technical documentation, workplace safety, and professional communication within a high-performance industrial organization.
- Developed problem-solving, teamwork, and analytical thinking skills by seeing real engineering practices and supporting tasks related to industrial material services and operational decision-making.

Projects:

Hybrid Thermal Battery Wall Panel Using Phase Change Material (PCM) – Senior Design Project Feb 2026 – May 2026

- My role was a leader which is leading a multidisciplinary senior design project focused on developing a PCM-based thermal battery wall panel to improve and control indoor thermal buffering and shift cooling loads.
- Integrating thermal analysis, airflow design, sensing, and low-voltage control using Phase Change Material (PCM).

Gas Oil Separation Plant (GOSP) – Process Simulation & Optimization July 2026 – Oct 2026

- Developed an integrated steady-state GOSP simulation using Aspen HYSYS, covering oil/gas/water separation, dehydration, desalting, crude stabilization, gas compression, and heat integration.
- Conducted sensitivity analysis on feed throughput, stabilizer reflux ratio, and separator pressure to evaluate production performance, product quality, and energy consumption.
- Performed energy and operating cost analysis, along with CO_2/H_2S corrosion-rate assessment using FREECORP 2.2 under assumed operating conditions.

AI Smart Reactor Design & Selection Model May 2026 - August 2026

- Developed an interactive reactor-design platform using Python and Streamlit for reactor sizing, performance evaluation, comparison, and engineering decision-making.
- Integrated chemical reaction engineering, mathematical modeling, reaction kinetics, Arrhenius inputs, data visualization, and an intelligent reactor recommendation workflow.
- Implemented CSTR, PFR, Batch Reactor, and Packed Bed Reactor models with liquid/gas-phase calculations, reversible/irreversible reactions, pressure-drop options, saved projects, and PDF engineering reports.

Controlling Microplastic Pollution from Polymeric Sports Field Materials Jan 2024 - May 2024

- Academic research report examining microplastic pollution generated from polymeric materials used in artificial sports fields, particularly rubber infill systems.
- Analyzed polymer-based microplastic release and evaluated engineering, alternative-material, and removal strategies to reduce environmental contamination.

Volunteering activities:

Assistant, Voluntary KFUPM Career Day at KFUPM Oct 2024 - Nov 2024

- Organized and coordinated KFUPM Career Day activities, supporting event logistics, communication, and student engagement with participating organizations.

Founder, Hydroxed Team Jun 2022 - Dec 2024

- Co-founded a creative team focused on motion graphics, graphic design, visual content, and client-based media services.
- Managed project planning, client communication, creative direction, and team coordination from concept development to final delivery.

Content Creator at KFUPM Media Club Oct 2021 - May 2022

- Produced digital content, motion graphics, and video media to support club campaigns, student events, and audience engagement.
- Supported event promotion and coordinated media content with club objectives and branding.

Professional Development:

Certifications:

- **Fundamentals of Process Simulation** – Aspen Technology
- **Upstream Oil and Gas industry** – KFUPM
- **Process Control System** – American Institute of Chemical Engineering
- **Finishing Process and Hydrotreating** – Saudi Aramco
- **Operator Qualification in Refining** – Saudi Aramco
- **Process Reactors** – Saudi Aramco

Skills:

Hard skills:
Chemical Reaction Engineering & Reactor Design | Python | Streamlit | MATLAB | Process Design & Simulation | Mathematical Modeling | Data Visualization | Polymer Materials & Non-Metallic Materials | Polymer Characterization | Fluid Mechanics | Technical Writing

Soft skills:
Problem-Solving | Analytical Thinking | Collaboration | Communication | Engineering Decision-Making | Time Management | Leadership & Delegation | Project Coordination | Client-Focused Mindset

Languages:

Arabic | English

References:

Available upon request.